Comparison technique for semilinear problems

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The Set Up

- Let (M^n, g) be a Riemannian manifold such that there exists $k \in \mathbb{R}$,

$$\mathrm{Ric}_g \geq k(n-1)g.$$

- Let $\Omega \subset M^n$ be a paracompact domain with C^2 boundary.
- Let $f \in Lip_{loc}(\mathbb{R})$ be a locally Lipschitz function such that

$$f(x) > 0 \quad \text{and} \quad f(x) \geq nkx + f(0) \quad \text{for } x \in I_f,$$

with $f(0) > 0$ when $k \leq 0$. Such that there exist non-empty connected intervals

$$0 \in R_f \subset [0, \bar{r}_k) \quad \text{and} \quad I_f \subset \mathbb{R}_+ \text{ with } 0 \in \partial I_f.$$

Then we consider the following **semilinear elliptic problem**,

$$\begin{cases} \Delta u + f(u) = 0 \text{ in } \Omega, \\ u > 0 \text{ in } \Omega, \\ u = 0 \text{ in } \partial\Omega. \end{cases} \quad (☠)$$

Basis for the Comparison

Model manifold

$$(M_k^n(F), g_k) := (I_k \times F, dr^2 + s_k(r)^2 g_F).$$

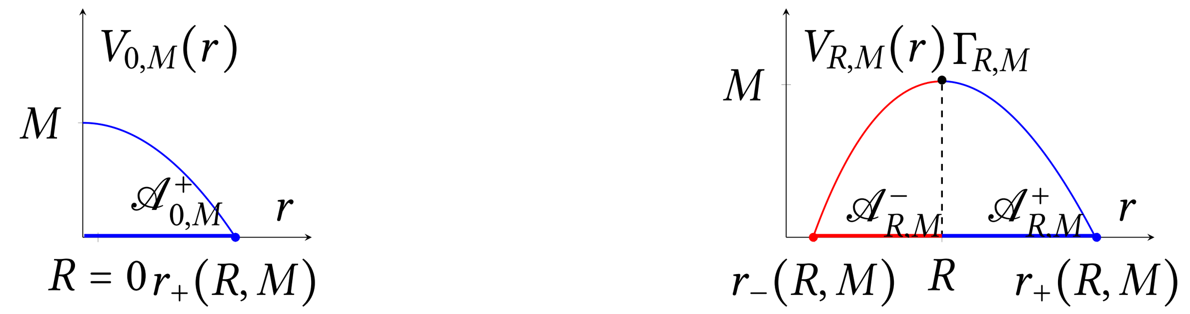
The Laplace–Beltrami operator on a model manifold,

$$\Delta = \partial_r^2 + (n-1)\cot_k(r)\,\partial_r + \frac{1}{s_k(r)^2}\Delta_F,$$

Model Solutions (*Radial solutions*)

$$v(p) = V(r(p))\ \forall p \in \mathscr{A} \Leftrightarrow \begin{cases} V''(r) + (n-1)\cot_k(r)V'(r) + f(V(r)) = 0, \\ V(R) = M, \qquad V'(R) = 0. \end{cases} \quad (☺)$$

For each initial data (R, M), we have two types of pairs solutions-domains, $(v_{R,M}, \mathscr{A}_{R,M})$, called **model pairs** and denote by $Model_M$,



$\mathscr{A}_{0,M} = \{ p \in M_k^n(F) : r(p) < r_+(0,M) \}$ $\qquad$ $\mathscr{A}_{R,M} = \{ p \in M_k^n(F) : r_-(R,M) < r(p) < r_+(R,M) \}$

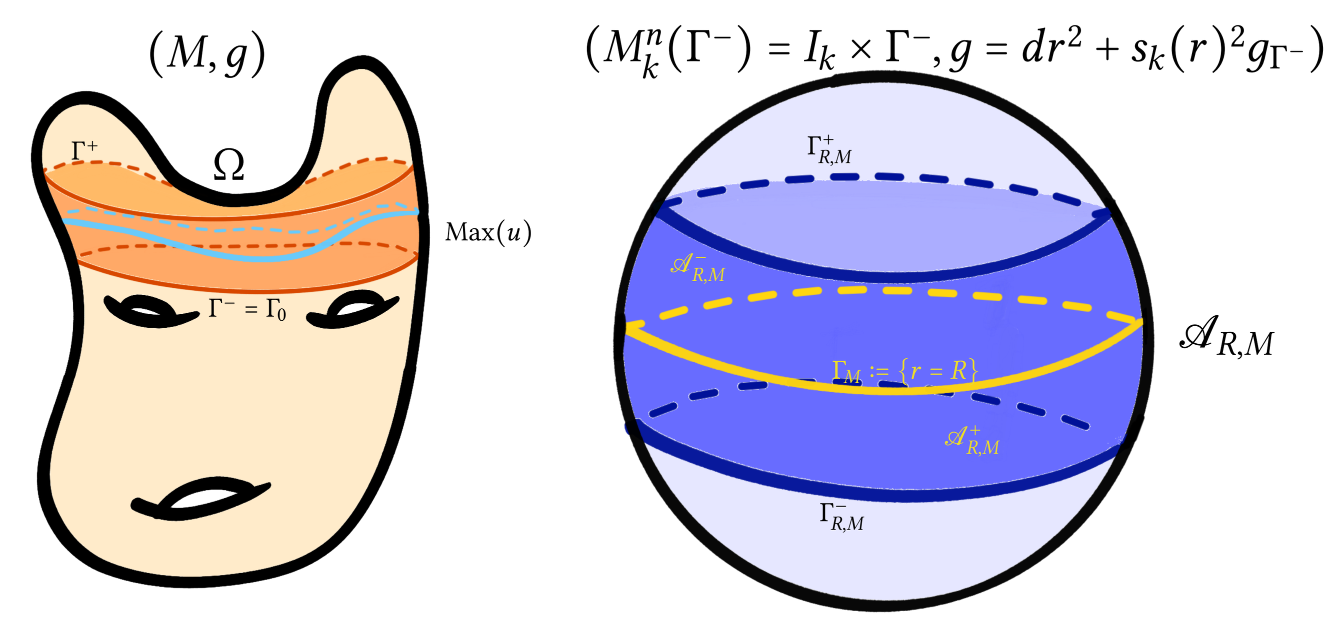
Model $\bar{\tau}$-functions are defined as,

$$\bar{\tau}_M^{\pm}(R) := \frac{V'_{R,M}\big(r_{\pm}(R,M)\big)^2}{V'_{0,M}\big(r_+(0,M)\big)^2}$$

Comparison algorithm

We fix $\mathfrak{U}$ be a connected component of $\Omega \setminus Max(u)$. For each connected component of the boundary $\Gamma \in \pi_0(\partial\mathfrak{U})$, define

$$\begin{cases} \bar{\tau}(\Gamma) := \max\limits_{p \in \Gamma} \dfrac{|\nabla u(p)|^2}{U'_{0,M}\big(r_+(0,M)\big)^2}, \\ \bar{\tau}(U) := \max\{\bar{\tau}(\Gamma) : \Gamma \in \pi_0(\partial\Omega \cap \mathrm{cl}(\mathfrak{U}))\} \\ \bar{\tau}(\Omega) := \max\{\bar{\tau}(\mathfrak{U}) : \mathfrak{U} \in \pi_0(\Omega \setminus Max(u))\}. \end{cases}$$

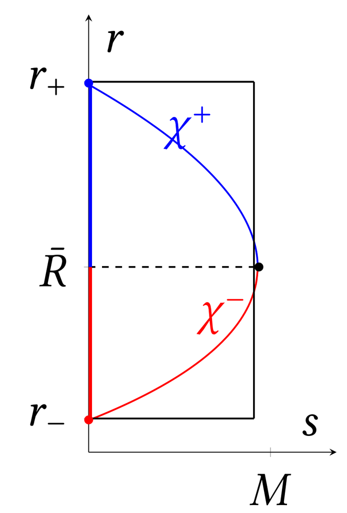


We say that a model pair in $M_k^n(\Gamma_0)$ $(\bar{\mathscr{A}}, \bar{u}) := (\mathscr{A}^{\pm}_{R,M}, v^{\pm}_{R,M}) \in Model^{\pm}_M$ is:

- A *comparison pair* for $(\mathfrak{U}, u)$ if $\bar{\tau}(\mathfrak{U}) \leq \bar{\tau}(\bar{\mathscr{A}}) = \bar{\tau}^{\pm}_M(R)$,
- An *associated model pair* if $\bar{\tau}(\mathscr{A}) = \bar{\tau}(\bar{\mathscr{A}}) = \bar{\tau}^{\pm}_M(R)$

We fix $(\bar{\mathscr{A}}, \bar{v})$ a comparison pair to $(\mathfrak{U}, u)$ with core radius $\bar{R}$ and zeroes in $\bar{r}_{\pm}$.

The comparison tool



We define the function,

$$G : [0, M] \times [\bar{r}_-, \bar{r}_+] \to \mathbb{R}$$
$$(s, r) \to G(s, r) := s - \bar{V}(r).$$

By the Implicit Function Theorem, there exist two C^2-functions, $\chi^{\pm} : [0, M) \to [\bar{r}_-, \bar{R}]$ or $[\bar{R}, \bar{r}_+]$, such that $G(s, \chi^{\pm}(s)) = 0$ for all $s \in [0, M)$, i.e. $\bar{V}(\chi^{\pm}(s)) = s$.

The *pseudo-radial function* associated to $(\mathfrak{U}, u)$ and $(\bar{\mathscr{A}}, \bar{v})$ is

$$\Psi : \mathfrak{U} \to \chi^{\pm}(0, M) \subset [\bar{r}_-, \bar{r}_+], \qquad \Psi(p) := \chi^{\pm}(u(p)),$$

which will allow us to compare the geometry of level sets.

Notation

Let $I_k := [0, \bar{r}_k)$, where $\bar{r}_k := +\infty$ if $k \leq 0$ and $\bar{r}_k := \pi/\sqrt{k}$ if $k > 0$.

$$s_k(r) := \begin{cases} \frac{\sinh(\sqrt{-k}r)}{\sqrt{-k}} & \text{if } k < 0 \\ r & \text{if } k = 0, \\ \frac{\sin(\sqrt{k}r)}{\sqrt{k}} & \text{if } k > 0, \end{cases} \quad \cot_k(r) := \frac{s'_k(r)}{s_k(r)} \quad \text{and} \quad \tan_k(r) = \frac{s_k(r)}{s'_k(r)}.$$

$r_\Omega = \max\{\mathrm{dist}(p, \partial\Omega) : p \in \Omega\}$

Comparison results

Gradient estimates

Let $f \in C^1(\mathbb{R})$ be a function satisfying condition:

$$f'(x) \geq nk \quad \text{for all } x \in I_f. \qquad (\bigstar)$$

We have

$$|\nabla u|^2(p) \leq \bar{V}'(\Psi)^2(p) \quad \text{for all } p \in \mathfrak{U}.$$

If equality holds at any point $p \in \mathfrak{U}$ $\begin{cases} 1.\ Ric(\nabla u, \nabla u) = (n-1)k|\nabla u|^2 \text{ on } \mathfrak{U}, \\ 2.\ (M^n, g) \text{ is isometric to } (\mathrm{M}_k^n(\Gamma_0), g_k) \text{ on } \mathfrak{U}, \\ 3.\ (\mathfrak{U}, u) \equiv (\bar{\mathscr{A}}, \bar{v}). \end{cases}$

Isoperimetric type inequality

Let $f \in C^1(\mathbb{R})$ the condition $(\bigstar)$ and

$$f(x) \leq f_k(x) = nkx + 1$$

Assume that Ω is contained in a convex domain of M and that $\bar{\tau}(\Omega) = \bar{\tau}(\mathfrak{U})$. If $k > 0$ suppose that $\bar{r}_+ \leq \bar{r}_k/2$. Then, for all $p \in \mathrm{Max}(u)$:

$$\frac{dist(p, \partial\Omega)}{\tan_k(r_\Omega)/n} \begin{cases} \geq \dfrac{\bar{r}_+ - \bar{R}}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (\bar{\mathscr{A}}, \bar{v})) \in Model^+_M, \\ > \dfrac{\bar{R} - \bar{r}_-}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (\bar{\mathscr{A}}, \bar{v}) \in Model^-_M. \end{cases}$$

If equality holds at a point of Ω $\begin{cases} 1.\ (M, g) \text{ isometric to the space form } \mathbb{M}^n(k), \\ 2.\ f(x) = nkx + 1, \\ 3.\ (\Omega, u) \text{ is a ball solution of Serrin's problem}. \end{cases}$

Isoperimetric type inequality

Let $f \in C^m(\mathbb{R})$, with $m = \max\{n-2, 3\}$, satisfying condition $(\bigstar)$. Assume that the $(n-1)$-dimensional part of $\mathrm{cl}(\mathfrak{U}) \cap \mathrm{Max}(u)$, denoted by Γ_M, consists of a possibly disconnected Lipschitz-continuous hypersurface. Hence,

$$\frac{H^n(\mathfrak{U})}{H^{n-1}(\Gamma_M)} \geq \begin{cases} \dfrac{H^n(\bar{\mathscr{A}})}{H^{n-1}(\bar{\Gamma}_M)} & \text{if } (\bar{\mathscr{A}}, \bar{v}) \in \mathrm{Model}^+_M, \\ \dfrac{H^n(\bar{\mathscr{A}})}{H^{n-1}(\bar{\Gamma}_M)} & \text{if } (\bar{\mathscr{A}}, \bar{v}) \in \mathrm{Model}^-_M. \end{cases}$$

The equality holds if and only if $(\mathfrak{U}, u) \equiv (\bar{\mathscr{A}}, \bar{v})$.

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